

1. A shepherd boy gets bored tending town's flock to have some fun he cries out "wolf" even though no wolf is sight. The villagers run to protect flock, but then they get really mad when they realize the boy was playing a joke identify TP, FP, FN,

Actual \ Predicted	Predicted	
	Wolf = yes	Wolf = No
Wolf = yes	TP Prediction is correct wolf came	FN Prediction went wrong wolf came
Wolf = No	FP Prediction went wrong wolf didn't come	TN prediction correct wolf didn't come

2. Assume there are 100 images, 30 of them depicts a cat the rest do not. AML model predicts the accuracy of a cat in 25 of 30 cat image. it also predicts absence of a cat in 50 of 70 no cat images identify TP, FP, FN, TN

Actual \ Predicted	Predicted	
	cat = yes	cat = No
cat = yes	25	5
cat = No	20	50
	45	55
		100

i) Precision

$$\frac{TP}{TP+FP} = \frac{25}{45} = 0.5$$

$$\text{Precision} = 0.555 = 55.5\%$$

ii) Recall:

$$\frac{TP}{TP+FN} = \frac{25}{25+5} = \frac{25}{30} = 0.833$$

$$\text{Recall} = 83.33\%$$

iii) F1 Score

$$\frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2 \times 55.5 \times 83.33}{55.5 + 83.33} = 66.69\%$$

3. An email ~~Service~~ ^{Service} provider wants to classify emails as spam or notspam using features

- contains "offer" (yes/no)
- contains "free" (yes/no)
- Email "length" (short/long)

Sol:-

i) Aprior probabilities

$$\text{Spam} = 3 \quad \text{Total} = 6$$

$$\text{Not spam} = 3$$

$$p(\text{spam}) = \frac{3}{6} = 0.5$$

$$p(\text{Not spam}) = \frac{3}{6} = 0.5$$

ii) Conditional probabilities

Spam

$$P(\text{offer} = \text{yes} | \text{spam}) = \frac{2}{3}$$

$$P(\text{offer} = \text{no} | \text{spam}) = \frac{1}{3}$$

$$P(\text{free} = \text{yes} | \text{spam}) = \frac{3}{3} = 1$$

$$P(\text{free} = \text{no} | \text{spam}) = \frac{0}{3} = 0$$

$$P(\text{length} = \text{short} | \text{spam}) = \frac{2}{3}$$

$$P(\text{length} = \text{long} | \text{spam}) = \frac{1}{3}$$

Not spam

$$P(\text{offer} = \text{yes} | \text{not spam}) = \frac{1}{3}$$

$$P(\text{offer} = \text{no} | \text{not spam}) = \frac{2}{3}$$

$$P(\text{free} = \text{yes} | \text{not spam}) = \frac{0}{3} = 0$$

$$P(\text{free} = \text{no} | \text{not spam}) = \frac{3}{3} = 1$$

$$P(\text{length} = \text{yes} | \text{not spam}) = \frac{1}{3}$$

$$P(\text{length} = \text{no} | \text{not spam}) = \frac{2}{3}$$

iii) Naive bayes:

Spam:

$$P(O = \text{yes}, \text{Fre} = \text{no}, \text{length} = \text{short})$$

$$= \frac{1}{2} \times \frac{2}{3} \times 0 \times \frac{2}{3}$$

$$= 0$$

Not spam:

$$P(\text{not spam}, \text{offer} = \text{yes}, \text{free} = \text{no}, \text{length} = \text{short})$$

$$= \frac{1}{2} \times \frac{1}{3} \times 1 \times \frac{1}{3}$$

$$= \frac{1}{18}$$

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